iv. Freeze drying: Removal of water from a frozen product by sublimation is called freeze drying. Freeze dried foods will be of superior quality with light and porous texture.



▲ Plate 6.8 Machine Dried Vadagam

D. Preservation by Irradiation

Food irradiation also known as cold sterilization is another preservation technique. The foods are bombarded by high energy rays called gamma rays or by fast moving electrons to kill bacteria, fungi and insects and in some cases to delay fruit ripening or prevent sprouting in onions and potatoes. The goal of irradiation is to kill the microorganisms and inactivate the enzymes without altering the food.



▲ Plate 6.9 Irradiated Foods

6.2.2 Chemical Methods

A. Salt or Brine

It is an ancient preservation technique. Food is treated with salt or strong salt solution. Salt causes high osmotic pressure and shrinking of cell, dehydrates foods and microbes by drawing out moisture.

B. High Concentration of Sugar

Sugar has the ability to bind water and make it unavailable for microbial growth. It reduces the solubility of oxygen in moisture, which is essential for the growth and multiplication of microorganisms. Apples, oranges, guavas, grapes and pineapples are suitable for making jams and jellies. The fruits should be just ripe because the pectin content is high in such fruits.



▲ Plate 6.10 Orange Squash

C. Using chemical preservatives

The chemicals when added interfere with the cell membrane of the microorganisms, their enzyme activity or their genetic mechanisms. They also act as antioxidants. The common chemical preservatives permitted are

1. Benzoic acid (including benzoates)

   Sodium benzoate is a salt of benzoic acid and is used in preservation of coloured fruit juices and squashes.

2. Sulphur dioxide (including sulphites):

   Potassium meta-bi-sulphite is used as a source of sulphur dioxide when it is

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